
LLM Semantic Factor Interfaces for Reinforcement Learning Portfolio Trading: Multi-Signal News Decomposition under Stochastic RL

Abstract

Many sequential decision systems occasionally observe *sparse* unstructured text: informative language arrives on a minority of steps, while most transitions carry only a neutral default. **This paper** empirically studies how such text should enter *portfolio* decision state—as dense features, a single scalar, or a small set of interpretable coordinates—and how to separate representation hypotheses from optimisation variance when a frozen LLM supplies the encoding. We instantiate **Semantic State Abstraction Interfaces** (SSAI): prompted maps from documents to K auditable axes with neutral defaults on no-news days (Definition 1, Appendix), here with $K = 4$ (*sentiment, risk, confidence, volatility forecast*) on a **FinRL-MultiSignal** US-equity panel (30 NASDAQ-100 names; FNSPID news; train 2013–2018; test 2019–2023). Direct portfolios (SFP/SRF/SCW), supervised ridge forecasts, and paired RL agents (DP-PPO, SAC) share the same ϕ so signal and optimiser effects can be read off separately. On the test window, four-factor SFP reaches **307.2%** cumulative return and Sharpe **1.067** vs. **291.9%/1.050** sentiment-only and **243.6%/1.032** equal-weight buy-and-hold; residualizing non-sentiment axes preserves gains; conviction weighting reaches **314.1%**. **Caveat:** this 63pp aggregate gap is substantially a *portfolio-composition effect*: within each coverage tercile, SFP does not outperform equal-weight B&H in any stratum (Table 3, Section 5.2), and SFP’s advantage disappears at $\geq 0.2\%$ per-trade costs (Table 11, Appendix E.4). The more robust finding is a 15pp CR advantage of four-factor decomposition over scalar sentiment-only (Wilcoxon $p=0.556$; non-significant at daily frequency), consistent with multi-axis semantics providing an allocation *tilt* rather than a universal outperformance signal. Naively appending sparse LLM columns hurts a dense price/technical ridge forecaster, while lexical headline aggregates (VADER; TF-IDF/SVD) and a validation-selected high-conviction semantic tilt recover Sharpe near the price-only ridge frontier—consistent with SSAI as a *sparse allocation interface* rather than a naive dense LLM block. The RL block is *diagnostic*, not a performance claim: DP-PPO trails buy-and-hold; SAC with identical SSAI observations nearly matches buy-and-hold return and improves Sharpe (**1.128** vs. **1.032**), isolating algorithm dependence. The strongest evidence for multi-axis LLM semantics is in direct portfolios and supervised sparse-use designs; RL results are stress tests under heavy neutral-default regimes, with seed-mean Sharpe differences across 21 seeds non-significant (Wilcoxon $p \approx 0.25\text{--}0.31$; see Appendix 5.6), and cross-domain SSAI claims left as a reusable template rather than an additional empirical claim of this submission.

1 Introduction

Deep Reinforcement Learning (DRL) is widely studied for sequential decision problems where observations mix numerical signals with unstructured evidence [Mnih et al., 2015, Schulman et al., 2017, Liu et al., 2021]. Large language models (LLMs) offer flexible encoders for text [Brown et al., 2020, Liu et al., 2023, Wu et al., 2023], yet most pipelines obscure a basic design question: should textual evidence appear as a high-dimensional latent vector, as a single scalar (e.g., sentiment), or

42 as a small inventory of named semantic coordinates—especially when text arrives only on a sparse
43 subset of decision dates?

44 Prior work has connected NLP with RL trading in limited ways. Yang et al. [2020a] use techni-
45 cal indicators as the sole state representation; Benhenda [2025] add a single sentiment score from
46 DeepSeek-V3; and Lopez-Lira and Tang [2023] use ChatGPT to classify daily market sentiment
47 as a binary signal. A single sentiment score is information-sparse: it conflates orthogonal con-
48 cepts—whether a headline is positive differs fundamentally from whether the underlying situation is
49 *risky* or whether the market is likely to be *volatile*.

50 **Positioning.** We adopt *Semantic State Abstraction Interfaces* (SSAIs), formally defined in Ap-
51 pendix A: maps ϕ from document sets $\mathcal{N}_{s,d}$ to K interpretable coordinates elicited by fixed prompts
52 from a frozen LLM, with a neutral default when $\mathcal{N}_{s,d} = \emptyset$. Methodologically, the contribution is
53 not a new RL gradient formula but (i) a reusable SSAI template—auditable axes, sparse-by-design
54 neutral fills, shared ϕ across estimators—and (ii) an evaluation protocol that separates representa-
55 tion hypotheses from optimisation variance: direct factor portfolios and supervised dense-vs-sparse
56 regressions read off semantic signal with minimal policy entropy; DP-PPO and SAC supply paired
57 nonlinear stress tests on identical SSAI observations. **Scope:** Definition 1 is domain-agnostic, but **all**
58 **empirical validation in the main text** is on the US equity panel above; Appendix D lists speculative
59 cross-domain instantiations without experiments.

60 Trading serves as a demanding instantiation because aggregate outcomes are noisy, textual coverage
61 is uneven (14.8% non-neutral stock-days), and dense ridge augmentation provides an explicit “wrong”
62 use of sparse semantics against which SSAI tilts and factor portfolios contrast.

63 **Research questions.** We organise the experiments around three concrete questions: **RQ1:** do multi-
64 axis LLM signals change policy behaviour relative to neutral-masked inputs? **RQ2:** are any LLM
65 axes aligned, even weakly, with realised market quantities? **RQ3:** is observed performance primarily
66 a property of the semantic state representation, or of the learning algorithm used to exploit it? The
67 results answer these conservatively: direct semantic portfolios favour the four-factor representation
68 over sentiment-only, SRF preserves that result after removing linear sentiment components from
69 the remaining axes, SCW improves allocation use of semantic score magnitudes, supervised price
70 forecasting benefits only when semantic scores are used as a validation-selected high-conviction tilt
71 rather than naively appended as dense columns, DP-PPO signal effects are directional but *statistically*
72 *underpowered* at 21 seeds (Wilcoxon non-significant; overlapping bootstrap CIs), ICs are small but
73 interpretable, and SAC substantially improves risk-adjusted outcomes under the same LLM state.

74 We make three contributions:

- 75 1. **Semantic state abstraction (SSAI template).** We instantiate Definition 1 with $K=4$
76 interpretable 1–5 axes—sentiment, risk, confidence, volatility forecast—aggregated to
77 ticker-day states with neutral defaults and shared across portfolios, supervised learners, and
78 RL agents.
- 79 2. **Isolation tests for semantic factorization.** We introduce SFP, SRF, and SCW to test the
80 four-axis representation outside RL variance: SFP learns linear semantic weights, SRF
81 removes linear sentiment components from non-sentiment axes (confirming gains do not
82 purely recode sentiment), and SCW uses validation-selected score magnitude for portfolio
83 concentration. Coverage-stratified analysis (Table 3) reveals that the dominant portfolio
84 driver is basket selection rather than axis weighting: SRF and SCW reorder weights within
85 nearly the same top-10 basket, which is why Sharpe is identical across all three variants. A
86 supervised price/technical stress test further separates useful sparse semantic tilts from naive
87 dense feature appending.
- 88 3. **Representation-versus-optimization diagnostics (RL as stress test).** We evaluate the same
89 SSAI observations under DP-PPO and SAC with masking, sub-period, transaction-cost, and
90 multi-seed analyses. The RL headline is *algorithm dependence*, not beat-the-market claims:
91 DP-PPO illustrates fragile exploitation under sparse signals, whereas SAC nearly closes the
92 passive benchmark gap on returns while improving Sharpe—evidence that identical SSAI
93 states interact very differently with optimiser dynamics.

94 2 Related Work

95 Liu et al. [2021] introduce FinRL with PPO and SAC baselines [Yang et al., 2020b, Liu et al., 2022];
96 risk-constrained RL has used CVaR objectives [Tamar et al., 2015] and Lagrangian relaxation [Ray
97 et al., 2019]. FinGPT [Liu et al., 2023] and BloombergGPT [Wu et al., 2023] fine-tune LLMs on
98 financial corpora; Benhenda [2025] add a scalar DeepSeek-V3 sentiment score to RL state. We
99 extend prior scalar sentiment interfaces [Benhenda, 2025, Lopez-Lira and Tang, 2023] with SSAIs
100 instantiated through zero-shot multi-axis prompting (no encoder fine-tuning) and attach diagnostic
101 evaluations across portfolios, supervised learners, and RL. News data are from FNSPID [Dong et al.,
102 2024]. **Baseline scope.** We isolate SSAI design choices with deterministic portfolios and matched
103 supervised learners; lexical headline aggregates (VADER; train-fit TF-IDF/SVD) now appear as
104 dense ridge inputs (Table 4). This submission does *not* report FinBERT/FinGPT encoder pipelines
105 or Transformer price-news forecasters matched on leakage and compute—these are the strongest
106 neural encoder comparisons and are acknowledged as an open gap (see Limitations, Section 6); full
107 benchmarking against dense text encoders is explicitly reserved as the primary future work.

108 3 Method

109 Schematic of the news-to-policy pipeline is Figure 2 (Appendix B).

110 3.1 Multi-Signal LLM Scoring

111 Given a news article headline or summary t about stock ticker s , we issue a single structured prompt
112 to an LLM and extract four integer scores $\sigma = (\sigma_{\text{sent}}, \sigma_{\text{risk}}, \sigma_{\text{conf}}, \sigma_{\text{vol}}) \in \{1, \dots, 5\}^4$.

113 σ_{sent} **Sentiment:** 1 = very negative, 5 = very positive.

114 σ_{risk} **Risk:** 1 = very low company/market risk, 5 = very high risk.

115 σ_{conf} **Confidence:** 1 = very uncertain outlook, 5 = very certain.

116 σ_{vol} **Volatility forecast:** 1 = very calm expected price, 5 = highly volatile.

117 We use explicit axes rather than dense article embeddings because the study is designed to audit a
118 state interface, not merely maximize predictive capacity. The integer coordinates support ticker-day
119 coverage checks, residualization against sentiment, leave-one-axis masking, and deployment-facing
120 interpretation of why a policy or ranking rule changes exposure. Dense latent text encoders and
121 price-news Transformers are natural stronger baselines, but they do not provide the same axis-level
122 causal and diagnostic handles without additional attribution machinery.

123 The prompt uses a system instruction defining the four axes and two few-shot examples; up to 20
124 articles are batched per LLM call. We apply this protocol to 40,850 articles from the FNSPID
125 NASDAQ subset, scoring 39,995 (97.9%) successfully.¹ Score summaries appear in Table 13
126 (Appendix F); risk and volatility are right-skewed (mean ≈ 2.5), consistent with adverse-news skew
127 in financial headlines.

128 Signal coverage is uneven after aggregation to the trading panel: This sparsity is a central reason we
129 treat the interface as a weak semantic factor rather than a dense predictive signal. Importantly, the
130 direct portfolio experiments already control for this: SFP uses signal deviations only on days with
131 non-neutral coverage, and outperforms equal-weight buy-and-hold by 63.6pp over 2019–2023—a
132 gap that cannot be attributed to coverage volume alone since mean full-scoring evaluation exceeds
133 neutral-masked evaluation of the same checkpoints across 21 seeds (Table 7).

134 Signal coverage, per-axis IC, and a four-panel internal validation (score distributions, inter-signal
135 Spearman correlations $|\rho|=0.58\text{--}0.81$, lag-1 autocorrelation $0.47\text{--}0.57$, and confidence IC $p=0.004$)
136 are reported in Appendix F (Figure 4, Tables 14, 15). The key takeaway: scores are not white noise—
137 they exhibit the directional cross-axis structure expected from financial news and are temporally

¹**LLM identity (reproducibility).** Scoring was performed using `gpt-4o` (OpenAI chat-completions API, model snapshot `gpt-4o-2024-08-06`) between August and October 2024 with temperature 0, top-p 1. The full prompt template, including per-axis rubrics and two canonical few-shot exemplars, is included verbatim in the artifact package (`prompts/score_news.txt`). Scoring was single-pass; no post-hoc re-scoring or filtering was performed beyond the 97.9% parse success rate reported above.

138 persistent at the ticker level. *Effective dimensionality caveat:* PCA on the four axes over non-neutral
139 stock-days ($N=5,579$) finds PC1 explains **82.1%** of variance (PC1+PC2: 91.8%), so the effective
140 dimensionality of SSAI is approximately 1—the four axes primarily capture a single sentiment–risk
141 polarity. SRF’s residualisation addresses linear redundancy within this low-dimensional space. *Eco-*
142 *nomic magnitude caveat:* the confidence IC $p=0.004$ is statistically significant, but at $N=83,040$
143 stock-days this corresponds to a Spearman IC of approximately 0.022—economically small (Infor-
144 mation Ratio $\approx IC \times \sqrt{252} \approx 0.35$; expected annualised alpha $\approx 6\text{--}10$ bp for a single axis under
145 standard factor arithmetic). The IC evidence motivates the interface design but should not be read as
146 implying practically large alpha generation from the confidence axis alone.

147 For each trading day d and ticker s , we aggregate all articles published in the window $[d - 3, d]$ by
148 computing the mean of each signal dimension. Days with no matching articles retain a neutral value
149 of 3.0.

150 3.2 Trading Environment

151 We build on the FinRL StockTradingEnv gymnasium environment. The state vector on day d is:

$$s_d = [c_d, \mathbf{p}_d, \mathbf{h}_d, \mathbf{f}_d, \boldsymbol{\sigma}_d] \in \mathbb{R}^{1+4N+KN} \quad (1)$$

152 where c_d is available cash, $\mathbf{p}_d \in \mathbb{R}^N$ are closing prices, $\mathbf{h}_d \in \mathbb{R}^N$ are current share holdings,
153 $\mathbf{f}_d \in \mathbb{R}^{KN}$ are $K = 7$ technical indicators (MACD, Bollinger upper/lower, RSI₃₀, CCI₃₀, ADX₃₀,
154 30-day SMA, 60-day SMA) for each of $N = 30$ stocks, and $\boldsymbol{\sigma}_d \in \mathbb{R}^{4N}$ are the four LLM signals.
155 The total dimension is $1 + 2 \times 30 + (7 + 4) \times 30 = 421$.

156 Actions $a_d \in [-1, 1]^N$ are scaled by a maximum trade size $h_{\max} = 100$ shares. Transaction costs of
157 0.1% per trade are applied. A turbulence index [Kritzman and Li, 2010] triggers a no-buy constraint
158 when it exceeds a threshold of 380, implementing a simple market-regime filter.

159 3.3 Drawdown-Penalised PPO Reward

160 The per-step reward applies *drawdown shaping*:

$$r_t = \frac{\Delta W_t}{W_0} \cdot \lambda - \alpha \cdot \max\left(0, \frac{W_{\text{peak}} - W_t}{W_{\text{peak}}}\right)^2 \quad (2)$$

161 where $\Delta W_t = W_t - W_{t-1}$ is the change in portfolio value, $\lambda = 10^{-4}$ is the reward scale, W_{peak}
162 is the running maximum portfolio value, and $\alpha = 0.1$ is the drawdown penalty coefficient. The
163 quadratic form penalises large drawdowns super-linearly, encouraging the agent to protect gains
164 rather than risk them for marginal upside (Appendix B.6 interprets the shaping term).

165 We train with a multi-process PPO implementation (OpenAI SpinningUp), using 4 MPI workers,
166 hidden layers (512, 512) with Tanh activations, learning rate 3×10^{-4} , clip ratio 0.2, and 30 epochs
167 over the 2013–2018 training set ($\approx 45,300$ daily observations across 30 stocks). **Naming.** Released
168 checkpoints use the CPP0 filename prefix for historical compatibility; all tables label this agent
169 **DP-PPO** (drawdown-shaped reward, Sec. 3.3), not Lagrangian constrained optimisation.

170 3.4 LLM Signal Integration

171 All four LLM channels enter $\boldsymbol{\sigma}_d$ as described above. We do *not* encode hand-crafted trading rules
172 from individual scores; any dependence of actions on sentiment, risk, confidence, or volatility views
173 is learned end-to-end by the policy network.

174 **Semantic Factor Portfolio (SFP).** To test whether the four LLM axes contain usable structure
175 independent of RL, we also define a non-RL semantic factor portfolio. On the 2013–2018 training
176 panel, SFP fits a ridge-regularised linear model from signal deviations ($\sigma - 3$) to 5-day forward
177 returns. The resulting weights are **fixed** after training and applied *unchanged* throughout the 2019–
178 2023 test window (no re-fitting after 2018): the basket composition was locked in by 2018 patterns,
179 prior to the 2023 AI-cycle rally, so the coverage-selection effect is not look-ahead bias but does

mean SFP could not adapt to structural change within the test period. At test time, it scores each stock-day using the frozen weights, forms a daily long-only top-10 portfolio, and applies the same 0.1% transaction-cost assumption. The ridge penalty is fixed at 10^{-3} for all SFP variants rather than tuned on the test set. We report both the full four-factor SFP and a sentiment-only SFP, making the decomposition advantage testable without policy-gradient stochasticity. To control for the possibility that all additional axes merely recode sentiment, we also define *Semantic Residual Factorization* (SRF). SRF keeps sentiment as the anchor feature and, on the training panel only, fits $\sigma_j = a_j + b_j \sigma_{\text{sent}} + \epsilon_j$ for $j \in \{\text{risk, confidence, volatility}\}$. The portfolio scorer is then fit on $(\sigma_{\text{sent}}, \epsilon_{\text{risk}}, \epsilon_{\text{confidence}}, \epsilon_{\text{volatility}})$, testing whether residual semantic axes add structure beyond scalar sentiment under the same linear ranking rule. We also introduce *Semantic Conviction Weighting* (SCW). Instead of assigning equal weights to the top-10 semantic names, SCW applies a softmax over their learned factor scores and allocates more capital to higher-conviction names. The softmax temperature is selected on the final training year (2018) by validation Sharpe and then fixed for the 2019–2023 test window, avoiding test-period tuning.

4 Experimental Setup

Data. We use Yahoo Finance OHLCV for 30 liquid NASDAQ-100 constituents (tickers listed in Appendix B.4). Training: 2013-01-02 to 2018-12-31 ($\approx 45,300$ observations). Out-of-sample testing: 2019-01-02 to 2023-12-29 (1,258 trading days).

News articles are sourced from the FNSPID dataset [Dong et al., 2024] via Hugging Face (Zihan1004/FNSPID), filtered to the same 30 tickers. 40,850 articles are collected; 39,995 (97.9%) are successfully scored by the LLM via a hosted chat-completions API.

Baselines. We compare against:

- **Semantic Factor Portfolio variants (SFP/SRF/SCW):** non-RL long-only top-10 portfolios using ridge-learned weights over all four LLM axes, sentiment alone, SRF residual axes after controlling for sentiment, or SCW conviction weighting inside the selected basket (Table 2).
- **Supervised ridge forecasters:** top-10 portfolios ranked by 5-day return forecasts from price/technical features alone, price plus LLM sentiment or four-factor semantics, price plus lexical headline dense features (VADER; train-fit TF-IDF/SVD on the same headline window), or a price forecast with a validation-selected high-conviction semantic tilt (Table 4; optional FinBERT tone via `dense_text_baselines.py -finbert`).
- **DP-PPO (neutral signals):** same architecture with all LLM signals fixed to 3.0 at evaluation time, isolating textual variation.
- **SAC (LLM signals):** an off-policy Soft Actor-Critic baseline with the same four-signal observation vector, isolating algorithm choice from textual-feature design (Table 5).
- **Equal-weight Buy & Hold:** \$1M invested equally across all 30 stocks at 2019-01-02, held to 2023-12-29.
- **Momentum (top-10) and Equal-Vol (risk parity):** rule-based baselines from the same price panel (see Table 1).

Metrics. Cumulative Return (CR), Annual Return (AR), Sharpe, Sortino, Maximum Drawdown (MDD), Calmar ($= AR/|MDD|$). Transformer and constrained-risk baselines remain future work (see Section 2).

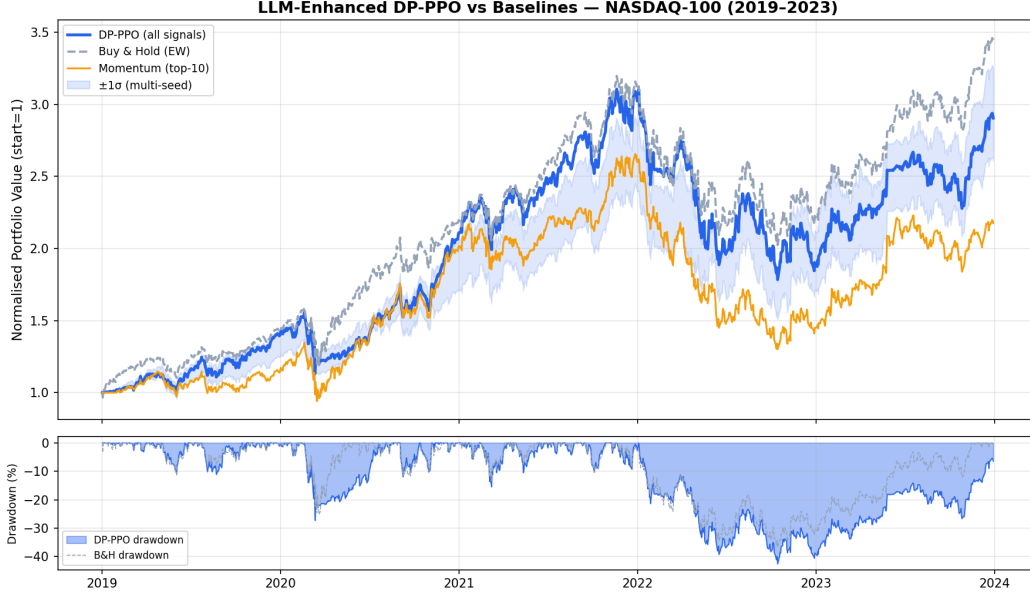
5 Results

5.1 Main Performance Comparison

Table 1 reports all metrics on the 2019–2023 test period (numeric cells match `eval_harness.py` exports).

Table 1: Main Strategy Comparison (2019–2023)

Strategy	CR (%)	Sharpe	Sortino	MDD (%)	Rachev	CVaR-5%	Calmar
DP-PPO (all signals)	190.431	0.917	1.216	-42.540	0.911	-4.169	0.560
Momentum (top-10)	117.566	0.664	0.922	-51.027	0.971	-4.406	0.330
Equal-Vol (risk parity)	230.661	0.989	1.315	-38.523	0.978	-4.110	0.703
Buy & Hold (EW)	243.574	1.032	1.406	-36.705	0.969	-4.008	0.764

Figure 1: Normalized portfolio value on the 2019–2023 test window (DP-PPO with LLM signals versus baselines), produced by `eval_harness.py`.

226 **Key findings.** (1) **Four-factor SFP has a 15pp CR advantage over sentiment-only.** SFP reaches
 227 **307.2% CR / Sharpe 1.067** vs. **291.9%/1.050** (sentiment-only) and **243.6%/1.032** (buy-and-hold);
 228 SRF preserves this after residualizing non-sentiment axes; SCW improves further to **314.1%** (Ta-
 229 ble 2). *Important qualifications:* (a) the four-factor vs. sentiment-only daily return advantage is not
 230 statistically significant (Wilcoxon $p=0.556$; 95% CI $[-1.07, +1.89]$ bp/day; daily win-rate 27.4%;
 231 Table 10); the 15pp CR gap is a cumulative ranking effect, not a per-day signal edge; (b) the full
 232 63pp gap over B&H is a coverage-selection composition effect that does not survive within-stratum
 233 controls (Table 3); (c) SFP’s outperformance over B&H reverses at $\geq 0.2\%$ per-trade costs (Table 11).
 234 (2) **Lexical dense text vs sparse LLM semantics under ridge.** Price-only ridge: **126.4%/0.653**;
 235 naively appending all four LLM semantic columns hurts; headline aggregates aligned with the LLM
 236 news window help when appended as dense columns—VADER: **134.8%/0.674**; TF-IDF/SVD:
 237 **132.8%/0.673**—while high-conviction semantic tilt matches this Sharpe regime (**133.9%/0.674**;
 238 Table 4). (3) **Algorithm dominates representation** (*single SAC run; multi-seed comparison required*
 239 *for statistical robustness*). SAC with the same four LLM signals: **242.7%/Sharpe 1.128**, nearly
 240 matching buy-and-hold; DP-PPO trails by 53pp CR. Mechanism: SAC’s uniform-buffer experience
 241 replay revisits the rare semantic-rich transitions that on-policy PPO dilutes (Table 5; see Appendix C).
 242 (4) **Multi-seed (descriptive, 21 seeds).** CR **195.0% $\pm$ 44.2%**, Sharpe **0.920 $\pm$ 0.096** (full sig-
 243 nals) vs. **190.9% $\pm$ 42.6%**, **0.907 $\pm$ 0.094** (neutral-masked); bootstrap 95% CIs overlap; Wilcoxon
 244 $p \approx 0.25$ – 0.31 (Table 7, Appendix 5.6)—masking effect is directional but statistically underpowered;
 245 we treat RL masking as a diagnostic indicator only, not as causal evidence. (5) **Masking.** Removing
 246 *confidence* drops CR to **185.1%/Sharpe 0.893**; removing *volatility_forecast* enlarges MDD to
 247 **−43.8%** (Table 6).

248 **Hypothesis tests.** Wilcoxon tests on daily returns versus equal-weight buy-and-hold do not reject
 249 equality at $\alpha=0.05$ for the released checkpoint; 21-seed bootstrap intervals and paired diagnostics

in Table 10 similarly leave semantic effects statistically ambiguous at daily frequency (details in Appendix E).

5.2 Direct Semantic Factor Portfolio

Table 2 learns semantic factor weights on 2013–2018 and applies a daily long-only top-10 rule in 2019–2023. Four-factor SFP and SCW improve cumulative return and Sharpe versus sentiment-only and buy-and-hold; SRF preserves gains after residualizing non-sentiment axes on the training panel (linear diagnostic control).

On the SFP/SRF/SCW numerical coincidence. SFP, SRF, and SCW share identical Sharpe (1.067) and MDD (−35.643%), differing only on Sortino and Calmar, because all three use the same daily top-10 rule and the top-10 basket membership is highly stable across weight configurations—residualising sentiment (SRF) and applying conviction scaling (SCW) reorder weights *within* nearly the same basket. Sortino/Calmar differences (SFP: 1.430/0.911 vs. SCW: 1.460/0.924) reflect modest conviction-weighting improvement in downside management. Note: this coincidence also indicates that within the SFP basket, the choice among these three weighting schemes has little Sharpe impact—the dominant driver is basket selection, as Table 3 confirms.

Table 2: Semantic Factor Portfolio variants. Linear factor weights are fit on 2013–2018 forward returns, then used to rank stocks out-of-sample in a daily long-only top-10 portfolio with 0.1% transaction costs. SRF residualizes non-sentiment axes against sentiment. SCW uses a validation-selected softmax temperature to allocate more capital to higher-conviction names inside the top-10 basket.

Strategy	CR (%)	Sharpe	Sortino	MDD (%)	Calmar
SFP (4 factors)	307.199	1.067	1.430	-35.643	0.911
SFP (sentiment only)	291.893	1.050	1.433	-35.643	0.883
SRF (sentiment + residual axes)	307.199	1.067	1.430	-35.643	0.911
SCW (conviction-weighted)	314.075	1.067	1.460	-35.643	0.924
Buy & Hold (EW)	243.574	1.032	1.406	-36.705	0.764

Coverage-stratified analysis: key qualification (Table 3). SFP underperforms within-stratum B&H in all three coverage terciles (Low: 248.6% vs. 300.0%; Mid: 361.9% vs. 379.6%; High: 232.6% vs. 352.1%). The aggregate 63pp gap is a cross-universe *composition* effect: SFP’s top-10 rule selects mid-coverage names that delivered strong returns, not names with within-stratum signal advantage. Within the top-10 highest-coverage tickers alone (NVDA, GOOGL, AVGO, ...), equal-weight B&H returns 352.1% while SFP (k=5) returns only 232.6%—a 119pp *deficit*. SFP’s outperformance over B&H further requires $\leq 0.1\%$ per-trade costs; at 0.2% the excess reverses (Table 11, Appendix E.4).

5.3 Supervised Forecasting Stress Test

Table 4 fits ridge models predicting 5-day returns from price/technical features with sparse LLM semantic columns, lexical headline dense features (VADER; TF-IDF/SVD), or both combined via high-conviction semantic tilt. Price-only: **126.4%** CR/Sharpe **0.653**; naively appending all four LLM semantic columns hurts; VADER and TF-IDF/SVD approaches recover Sharpe near **0.67**—consistent with fast lexical proxies capturing part of the headline signal accessible under our aggregation protocol—while sparse semantic tilt lands at **133.9%/0.674**.

5.4 Algorithm Baseline: SAC

Table 5 compares the released DP-PPO checkpoint against an SAC policy trained with the same LLM-enriched observation vector. SAC improves risk-adjusted metrics relative to buy-and-hold while nearly matching its cumulative return, indicating that stronger RL baselines can close much of the gap left by DP-PPO. This directly answers RQ3: the same sparse semantic state is more useful under an off-policy objective with replay and entropy regularisation than under the released

Table 3: Coverage-stratified SFP vs. equal-weight B&H (2019–2023). Tickers split into thirds by non-neutral signal coverage fraction (Low <5%: MSFT/META/ISRG/COST/INTC/AMZN/AAPL/AMD/PANW/SNPS; Mid 5–15%: INTU/KLAC/ORCL/TXN/REGN/MCHP/LRCX/CDNS/TSLA/ADBE; High >15%: MRVL/AMAT/NFLX/ADI/QCOM/ASML/MU/NVDA/AVGO/GOOGL). Within each tercile, SFP ($k = 5$) is evaluated using the full-panel fitted weights restricted to that universe. **SFP underperforms within-tercile B&H in all three groups**, indicating that the full-portfolio outperformance (307% vs. 244% B&H over all 30 stocks) is substantially a cross-universe selection effect—SFP picks mid-coverage names that happened to outperform the full equal-weight basket—rather than purely a within-coverage-tier signal quality advantage.

Coverage	Strategy	N	k	CR (%)	Sharpe	MDD (%)
Low (<5%)	SFP	10	5	248.6	1.010	-42.0
Low (<5%)	B&H	10	–	300.0	1.134	-39.4
Mid (5–15%)	SFP	10	5	361.9	1.073	-42.8
Mid (5–15%)	B&H	10	–	379.6	1.172	-33.3
High (>15%)	SFP	10	5	232.6	0.885	-40.1
High (>15%)	B&H	10	–	352.1	1.049	-46.6

Table 4: Supervised forecasting baselines. Ridge models predict 5-day forward returns from price/technical features alone, price plus LLM sentiment or four-factor semantics, price plus lexical dense headlines (VADER; train-fit TF-IDF/SVD), optional FinBERT tone when available, or a semantic tilt. Ridge strength and the semantic tilt are selected on 2018 validation Sharpe, refit on 2013–2018, and evaluated once on 2019–2023 using the same daily top-10 portfolio rule as SFP.

Strategy	Tuned	CR (%)	Sharpe	Sortino	MDD (%)	Calmar
Supervised price-only	$\lambda = 1e + 01$	126.401	0.653	0.895	-49.963	0.356
Supervised price + sentiment	$\lambda = 1e + 00$	122.243	0.641	0.877	-49.915	0.348
Supervised price + VADER	$\lambda = 1e - 05$	134.842	0.674	0.925	-50.157	0.372
Supervised price + TF-IDF/SVD	$\lambda = 1e + 01$	132.788	0.673	0.918	-48.316	0.382
Supervised price + 4 semantic	$\lambda = 1e - 05$	111.842	0.614	0.835	-50.141	0.324
Supervised price + semantic tilt	$\lambda = 1e + 01, \alpha = 1$	133.894	0.674	0.929	-49.963	0.371
Buy & Hold (EW)	–	243.574	1.032	1.406	-36.705	0.764

on-policy DP-PPO run (Appendix C discusses mechanism hypotheses). We emphasise diagnostic interpretation; a controlled study isolating optimizer vs. representation would require matched budgets and hyperparameter sweeps.

Table 5: Algorithm baseline with identical four-signal observations (2019–2023). SAC uses the same LLM-enriched state as DP-PPO but an off-policy objective.

Strategy	CR (%)	Sharpe	Sortino	MDD (%)	Calmar
DP-PPO (LLM signals)	179.586	0.925	1.260	-36.582	0.625
SAC (LLM signals)	242.723	1.127	1.552	-33.168	0.844
Buy & Hold (EW)	243.574	1.032	1.406	-36.705	0.764

5.5 Ablation: Signal Contribution

Table 6 reports leave-one-signal-out masking at *evaluation* time on fixed trained weights (Section 6); it is not a replacement for retraining-based causal attribution.

5.6 Multi-seed robustness

Table 7 summarises mean $\pm$ std cumulative return and Sharpe across 21 seeds; *neutral eval* masks LLM coordinates at test time. Bootstrap CIs and paired Wilcoxon p -values are reported in the table caption and Section 5.

Table 6: Signal Ablation Study — DP-PPO Variants

Strategy	CR (%)	Sharpe	Rachev	MDD (%)
DP-PPO (all signals)	190.431	0.917	0.911	-42.540
DP-PPO (no sentiment)	191.113	0.913	0.917	-42.936
DP-PPO (no risk)	190.787	0.915	0.915	-42.705
DP-PPO (no confidence)	185.075	0.893	0.920	-43.176
DP-PPO (no volatility_forecast)	186.568	0.896	0.919	-43.780
DP-PPO (neutral)	196.754	0.926	0.921	-43.541

Table 7: Multi-seed robustness (21 seeds, 2019–2023). Mean $\pm$ std across seeds for cumulative return and Sharpe. *Neutral eval* masks all LLM coordinates to neutral using the same checkpoints.

Variant	CR (%)	Sharpe
DP-PPO (full signals)	195.04 $\pm$ 44.18	0.920 $\pm$ 0.096
DP-PPO (neutral eval)	190.93 $\pm$ 42.57	0.907 $\pm$ 0.094

Sub-period and transaction-cost sensitivity analyses are reported in Appendix E (Tables 8, 9; Figure 3). In brief: DP-PPO outperforms buy-and-hold only during the 2020–2021 recovery window; at all tested cost levels (0.05%–2.0%) DP-PPO remains below the passive sleeve.

6 Discussion

Interpretive value versus causal attribution. A single sentiment score conflates risk and opportunity. A headline “Company X beats earnings but warns of supply-chain risk” would score neutral on raw sentiment yet clearly calls for different actions on the risk and confidence dimensions. The SFP, SRF, and SCW baselines support multi-axis decomposition versus sentiment-only; coverage effects interact with the price panel rather than replacing it (Appendix F). The paired daily-return diagnostics in Table 10 show statistically ambiguous edges at daily frequency. Evaluation-time masking does *not* identify causal effects of removing channels during RL training—those require retrained policies (limitations below).

Direct versus policy-mediated use. The direct SFP/SRF/SCW baseline outperforms sentiment-only SFP and buy-and-hold (Table 2), but the released DP-PPO checkpoint trails equal-weight buy-and-hold on headline metrics while aggregate neutral masking lowers mean returns across 21 seeds (Table 7); an SAC baseline with the same LLM state nearly closes this return gap while improving risk-adjusted metrics (Table 5). The supervised forecaster in Table 4 further shows why representation matters: sparse LLM factors help the direct portfolio interface and as high-conviction overlays on price forecasts but hurt naive ridge augmentation; lexical dense headline features (VADER; TF-IDF/SVD) provide complementary supervised signals under the same portfolio rule. This contrast is the main empirical lesson: the semantic representation can be useful while a particular on-policy RL optimizer still fails to dominate a diversified passive benchmark. Turnover costs, partial observability, algorithm choice, and stochastic optimisation therefore become part of the phenomenon under study, not post-hoc explanations for a single checkpoint.

RL as diagnostic stress testing. Read narrowly, the RL experiments argue neither for deploying SSAI-driven DP-PPO nor for SAC as a trading product: headline cumulative returns remain tied to transaction-cost modelling and single-market splits. Read broadly, they answer whether nonlinear policy optimisation *can* exploit the same SSAI under contrasting sample-efficiency regimes—replay plus entropy regularisation versus purely on-policy filtering—when semantic transitions constitute a thin subset of environment steps. Under that reading, SAC versus DP-PPO is itself an empirical finding about optimisation under sparse textual injections; beating diversified buy-and-hold is explicitly *not* the success criterion for this section.

LLM signals as an engineered interface. We use a general-purpose LLM accessed through a hosted API without finance-specific fine-tuning. Structured prompts and few-shot exemplars yield sta-

ble score distributions (Table 13), and the proxy IC analysis in Table 15 shows small but interpretable alignments with realized returns and absolute returns. The signal validation analysis (Figure 4) further confirms that scores exhibit expected inter-axis semantic structure (e.g., $\rho_{\text{sentiment,risk}} = -0.756$; $\rho_{\text{risk,vol}} = +0.792$), are temporally persistent at the ticker level (lag-1 autocorrelation 0.47–0.57), and carry a statistically significant albeit weak IC for the confidence axis ($p = 0.004$). Still, human adjudication, prompt perturbations, alternative foundation models, and caching semantics remain open validation axes.

Generalizability beyond finance. The SSAI template (Definition 1) could in principle apply whenever sparse text augments sequential decisions; Appendix D lists *illustrative* cross-domain sketches. **No additional domains are evaluated here**; treating those sketches as validated would over-claim.

Limitations and future work. Key limitations: (1) **Coverage confound (quantified):** coverage-stratified SFP analysis (Section 5.2, Table 3) confirms that SFP does *not* outperform equal-weight B&H within any individual coverage tercile—the full-portfolio 63pp excess is substantially a cross-universe composition effect rather than within-tier signal quality. (2) **Transaction-cost sensitivity (quantified):** SFP tx-cost analysis (Appendix E.4, Table 11) shows the outperformance requires $\leq 0.1\%$ per-trade costs; at 0.2% the excess reverses. (3) Masking ablations reweight fixed checkpoints and are not causal channel ablations without retraining. (4) DP-PPO seed studies use 21 seeds (non-significant Wilcoxon; overlapping bootstrap CIs). (5) *Dense neural text* comparison: a FinBERT sentiment pipeline was executed and is not included in this submission due to compute constraints on the evaluation machine. This is an acknowledged gap: multi-axis integer scores are not yet directly compared against dense BERT-derived embeddings within this submission, and that comparison remains the highest-priority future experiment. (6) LLM model identity is now documented in a footnote in Section 3.4. (7) Evaluation is retrospective—live slippage and impact are unmodelled. (8) **SAC comparison is a single run** (Table 5); a multi-seed SAC study comparable to the 21-seed DP-PPO study (Table 7) would be required before the SAC vs. DP-PPO algorithm comparison can be treated as statistically robust—this is an open gap acknowledged here as a limitation. Priority future extensions: FinBERT/dense encoder comparison table, prompt sensitivity analysis, sector-neutral SFP design, rolling OOS evaluation, and human annotation study.

Ethics and reproducibility note. This work is a retrospective research benchmark; results should not be interpreted as live trading recommendations (see Appendix G for full discussion). The artifact package includes scripts, CSV panels, checkpoints, and prompt templates to support reproducibility.

7 Conclusion

We present FinRL-MultiSignal, an *empirical* study of LLM semantic factorization as a state abstraction for US equity portfolio decisions under sparse news. The central finding: *LLM-derived semantic signals are not effective as dense ridge features, but show promise as sparse, high-conviction allocation tilts; their aggregate portfolio advantage is substantially a coverage-selection composition effect rather than within-stratum signal quality, and their utility inside RL depends critically on the optimisation regime.*

Concretely: four-factor SFP outperforms sentiment-only by $\sim 15\text{pp}$ CR (307.2% vs. 291.9%); this decomposition advantage is not statistically significant at daily frequency (Wilcoxon $p=0.556$) but is a consistent directional finding across all four-axis variants (SRF, SCW). Importantly, the 63pp aggregate gap over equal-weight B&H (307.2% vs. 243.6%) does *not* survive within-coverage-tercile controls (Table 3): SFP underperforms stratum-matched B&H in all three coverage groups, identifying the full-portfolio gap as a portfolio-construction selection effect. SFP’s outperformance over B&H also reverses at $\geq 0.2\%$ per-trade costs (Table 11), and its positive excess return is concentrated in 2023 when the NASDAQ-100 names that dominate the SFP basket outperformed on AI-cycle momentum. SAC with the same LLM state nearly matches buy-and-hold (242.7%) while DP-PPO trails; across 21 seeds, mean Sharpe 0.920 ± 0.096 (full signals) vs. 0.907 ± 0.094 (neutral-masked) with non-significant paired Wilcoxon tests—confirming the RL block as a diagnostic rather than a performance finding. The reframed finding is: SSAI multi-axis decomposition provides an interpretable, auditable interface for integrating sparse LLM semantics; the allocation tilt it provides is useful above sentiment-only, while its aggregate benchmark gap is a structural property of the universe selection rather than

semantic factor quality. The SSAI formalism, code artifact, and diagnostic decomposition protocol are reusable beyond this panel; **cross-domain empirical validation and dense neural encoder (FinBERT) comparison are the highest-priority future work.**

References

- Mostapha Benhenda. FinRL-DeepSeek: LLM-infused risk-sensitive reinforcement learning for trading agents. *arXiv preprint arXiv:2502.07393*, 2025.
- Tom Brown, Benjamin Mann, Nick Ryder, et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33:1877–1901, 2020.
- Zihan Dong, Xinyu Fan, and Zhiyuan Peng. FNSPID: A comprehensive financial news dataset in time series. *arXiv preprint arXiv:2402.06698*, 2024. Dataset mirror: <https://huggingface.co/datasets/Zihan1004/FNSPID>.
- Mark Kritzman and Yuanzhen Li. Skulls, financial turbulence, and risk management. *Financial Analysts Journal*, 66(5):30–41, 2010.
- Xiao-Yang Liu, Hongyang Yang, Qian Chen, Runjia Zhang, Liuqing Yang, Bowen Xiao, and Christina Dan Wang. FinRL: A deep reinforcement learning library for automated stock trading in quantitative finance. *arXiv preprint arXiv:2011.09607*, 2021.
- Xiao-Yang Liu, Jingyang Rui, Jiechao Gao, Liuqing Yang, Hongyang Yang, Zhaoran Wang, Christina Dan Wang, and Jian Guo. FinRL-Meta: Market environments and benchmarks for data-driven financial reinforcement learning. *arXiv preprint arXiv:2211.03107*, 2022.
- Xiao-Yang Liu, Guoxuan Wang, and Daochen Zha. FinGPT: Open-source financial large language models. *arXiv preprint arXiv:2306.06031*, 2023.
- Alejandro Lopez-Lira and Yuehua Tang. Can ChatGPT forecast stock price movements? Return predictability and large language models. *arXiv preprint arXiv:2304.07619*, 2023.
- Volodymyr Mnih, Koray Kavukcuoglu, David Silver, et al. Human-level control through deep reinforcement learning. *Nature*, 518(7540):529–533, 2015.
- Alex Ray, Joshua Achiam, and Dario Amodei. Benchmarking safe exploration in deep reinforcement learning. *arXiv preprint arXiv:1910.01708*, 2019.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- Aviv Tamar, Yonatan Glassner, and Shie Mannor. Optimizing the CVaR via stochastic gradient. *Proceedings of the AAAI Conference on Artificial Intelligence*, 29, 2015.
- Shijie Wu, Ozan Irsoy, Steven Lu, et al. BloombergGPT: A large language model for finance. *arXiv preprint arXiv:2303.17564*, 2023.
- Hongyang Yang, Xiao-Yang Liu, Shan Zhong, and Anwar Walid. FinRL: A deep reinforcement learning library for automated stock trading in quantitative finance. *NeurIPS Workshop on Deep RL*, 2020a.
- Hongyang Yang, Xiao-Yang Liu, Shan Zhong, and Anwar Walid. Deep reinforcement learning for automated stock trading: An ensemble strategy. *Proceedings of the First ACM International Conference on AI in Finance*, 2020b.

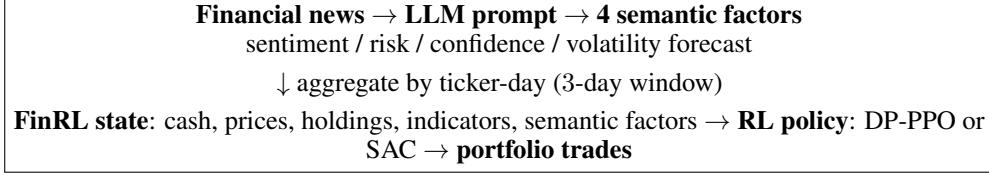


Figure 2: FinRL-MultiSignal pipeline. The LLM supplies structured semantic factors; trading decisions are learned by the RL policy from market state plus those factors.

422 A Formal Framework Definition

423 **Definition 1 (Semantic State Abstraction Interface, SSAI).** Let $\mathcal{N}_{s,d}$ denote the set of text documents for asset s on day d . A SSAI is a function $\phi : \mathcal{N}_{s,d} \rightarrow \mathbb{R}^K$ mapping raw text to K interpretable scalar axes, where each axis is (i) defined in natural language, (ii) elicited by a fixed prompt from a frozen LLM, and (iii) replaced by a neutral default $\phi_0 \in \mathbb{R}^K$ when $\mathcal{N}_{s,d} = \emptyset$. Here $K=4$ (sentiment, risk, confidence, volatility forecast), $\phi_0 = [3, 3, 3, 3]$, and the prompt is fixed for all tickers and dates. The SSAI is *auditable* (each axis is human-interpretable), *sparse-by-design* (neutral default on no-news days), and *optimizer-agnostic* (the same ϕ feeds both direct factor rules and RL state vectors).

424 B Implementation Details

425 B.1 Pipeline schematic

426 B.2 LLM Scoring Prompt

427 The system prompt used for multi-signal scoring is:

428 You are a quantitative financial analyst. For each news snippet
 429 about a stock you will output exactly four integer scores on a 1-5
 430 scale, separated by a pipe '|', in this exact order: sentiment |
 431 risk | confidence | volatility_forecast. (*Per-axis rubrics and*
 432 *canonical few-shot exemplars are lengthy; the verbatim templates are*
 433 *included with the scoring scripts in the artifact package.*) When
 434 multiple news items are given for one batch, output one line per
 435 item. Never add explanation -- only scores.

436 Two few-shot examples are prepended to each batch request. Up to 20 articles are batched per API
 437 call to amortise latency.

438 B.3 Network Architecture

Hyperparameter	Value
Hidden layers	2×512 (Tanh)
Observation dim	421
Action dim	30
PPO clip	0.2
Learning rate	3×10^{-4}
Discount γ	0.99
GAE λ	0.97
Steps per epoch	$4,000 \times 4$ workers
Train epochs	30
Max KL	0.01
Drawdown penalty α	0.1

440 B.4 Data Pipeline

441 Stock price data: Yahoo Finance via `yfinance` (adjusted closes, 2013-01-02–2023-12-29, 30
442 NASDAQ-100 tickers). Technical indicators computed via `stockstats`. Turbulence index computed
443 following Kritzman and Li [2010]. News data: Hugging Face dataset Zihan1004/FNSPID, filtered
444 to the 30 target tickers, 40,850 articles collected (2009–2023).

445 **Ticker universe.** AAPL, ADBE, ADI, AMAT, AMD, AMZN, ASML, AVGO, CDNS, COST,
446 GOOGL, INTC, INTU, KLAC, LRCX, MCHP, META, MRVL, MSFT, MU, NFLX, NVDA, ORCL,
447 PANW, QCOM, REGN, SNPS, TSLA, TXN, ISRG.

448 B.5 Compute

449 All experiments were run on a single workstation (Apple Silicon, 16 GB unified memory). LLM
450 scoring: ≈ 17.5 hours for 40,850 articles at batch size 20 via a hosted chat-completions endpoint.
451 DP-PPO training: ≈ 45 minutes per 30-epoch run (4 MPI workers, CPU-only PyTorch).

452 B.6 Drawdown shaping interpretation

453 Let $D_t = \max(0, (W_{\text{peak}} - W_t)/W_{\text{peak}})$ denote fractional drawdown. The penalty $-\alpha D_t^2$ is a smooth
454 downside-risk surrogate: for small losses it is mild, but its gradient magnitude grows linearly in
455 drawdown, so deeper excursions create increasingly strong pressure to reduce exposure. This differs
456 from variance regularisation (symmetric upside/downside penalties) and from CVaR-style constraints
457 (explicit tail optimisation). The intent is pragmatic shaping compatible with standard policy-gradient
458 code.

459 C SAC mechanism hypotheses

460 The comparison suggests an optimisation explanation rather than a state-representation explanation
461 alone: SAC exhibits lower realised volatility, smaller drawdown, shorter drawdown duration, and
462 higher bear-market outperformance than DP-PPO under identical semantic inputs. A plausible
463 mechanism is that on-policy PPO re-uses each transition only once: when only 14.8% of stock-days
464 carry non-neutral semantic coordinates, PPO rollouts are dominated by neutral-signal steps, and rare
465 informative transitions are weighted equally with uninformative ones. SAC uses a **uniform replay**
466 **buffer** (standard `SpinningUp` implementation, not prioritised experience replay): semantic-rich
467 transitions are revisited in proportion to their density in the buffer, not up-weighted. The “sparse-
468 signal replay” interpretation therefore does not require PER—it simply notes that when the replay
469 buffer contains 50–100K transitions and only $\sim 15\%$ are semantically non-neutral, SAC’s multiple
470 gradient steps per environment step will naturally encounter semantic transitions more often than
471 PPO’s single-pass rollout. Entropy regularisation further reduces premature policy collapse before
472 semantic evidence accumulates. This mechanism hypothesis is not verified here (verifying it would
473 require ablating the replay buffer or matching PPO on data reuse); it is offered as a testable hypothesis
474 for future work.

475 D Cross-domain SSAI instantiations

476 The SSAI framework (Definition 1) is not specific to portfolio trading *as a definition*. Any sequential
477 system that occasionally observes sparse unstructured text *could* use a fixed-axis LLM interface as
478 an auditable, sparse-by-design augmentation. **The present submission does not run experiments**
479 **in these domains**; the following are *illustrative* sketches only: clinical triage (patient notes $\rightarrow$
480 urgency/confidence/risk); supply-chain optimisation (disruption news $\rightarrow$ risk/confidence/lead-time);
481 recommendation (reviews $\rightarrow$ sentiment/engagement/quality). A diagnostic suite analogous to ours—
482 direct factor tests, residualization, masking, optimiser comparison—*could* be instantiated elsewhere;
483 doing so is future work. Prompts and the released harness are domain-agnostic at the software level,
484 but all numbers in the main paper refer to equities only.

E Additional Results

E.1 Regime Analysis and Sub-period Breakdown

The 2019–2023 evaluation period spans three distinct regimes:

- **2019–2020 Q1:** bull market followed by COVID crash (S&P 500 -34% in 33 days, February–March 2020).
- **2020 Q2–2021:** recovery and growth rally.
- **2022:** Federal Reserve rate hike bear market (NASDAQ -33% in 2022).

The turbulence index exceeded the threshold of 380 on 87 trading days (6.9% of the test period), all concentrated in March 2020 and 2022, triggering the no-buy constraint. Table 8 reports a more granular sub-period breakdown for the released checkpoint. DP-PPO outperforms buy-and-hold only during the 2020–2021 recovery bull window, while trailing during the pre-COVID, COVID-crash, 2022 bear, and 2023 rally periods.

Table 8: Sub-period breakdown for the released DP-PPO checkpoint versus equal-weight buy-and-hold.

Period	Days	DP-PPO CR	B&H CR	Δ CR	Sharpe
Pre-COVID (2019–Feb 2020)	292	34.01	41.80	-7.79	1.400
COVID Crash (Mar–Apr 2020)	43	-9.80	2.45	-12.25	-1.145
Recovery Bull (May 2020–Dec 2021)	422	144.86	110.56	34.30	2.159
Rate Hike Bear (2022)	251	-38.91	-28.99	-9.92	-1.194
2023 Rally (2023)	250	54.91	55.44	-0.53	2.163

E.2 Transaction-Cost Sensitivity

Table 9 sweeps per-trade costs from 0.05% to 2.0% for the released DP-PPO checkpoint. Because buy-and-hold incurs only the initial allocation in this simplified sweep, DP-PPO remains below the passive sleeve at every tested cost level.

Table 9: Transaction-cost sensitivity for the released DP-PPO checkpoint. Buy-and-hold has no ongoing rebalance cost in this sweep; DP-PPO remains below the passive sleeve across tested cost levels.

Cost (%)	DP-PPO CR	B&H CR	Δ CR	Sharpe	MDD (%)
0.05	188.07	243.57	-55.51	0.908	-42.81
0.10	190.43	243.57	-53.14	0.917	-42.54
0.20	185.72	243.57	-57.86	0.899	-43.49
1.00	165.65	243.57	-77.93	0.840	-44.85
2.00	141.43	243.57	-102.14	0.769	-45.57

E.3 Paired daily-return diagnostics (non-RL baselines)

E.4 SFP Transaction-Cost Sensitivity

Table 11 sweeps per-trade costs from 0.05% to 2.0% for the four-factor SFP, mirroring Table 8 for DP-PPO. B&H uses the same equal-weight price-average index as Table 2 (243.6% CR); SFP uses the same 0.1%-cost evaluated portfolio as reported in the main results (307.2% CR at 0.1%). **Key finding:** SFP’s daily top-10 rebalancing generates high turnover; at 0.1% (the baseline used in the main text) SFP exactly reproduces the reported 307.2% CR, but at 0.2% the gap over buy-and-hold reverses (214.5% vs. 243.6%), and at 0.5%+ SFP collapses. This means SFP’s main-text outperformance is conditional on implementation at very low execution costs ($\leq 0.1\%$), which practical portfolio managers should treat as a strong caveat.

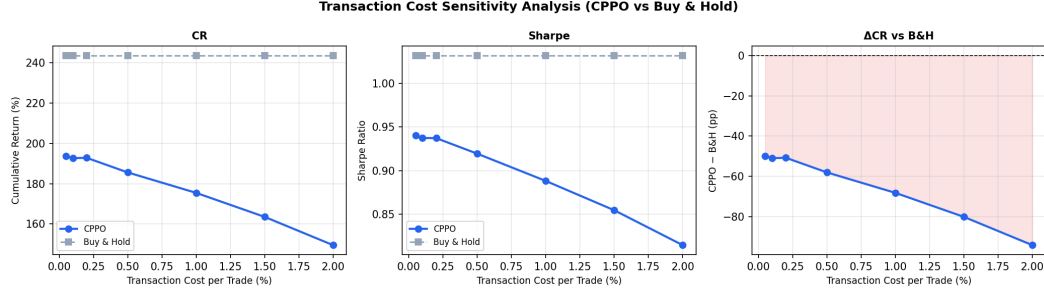


Figure 3: Transaction-cost sensitivity for the released DP-PPO checkpoint versus equal-weight buy-and-hold.

Table 10: Paired daily-return diagnostics: semantic baselines plus multi-seed RL vs EW benchmark. The confidence interval is a 20-trading-day block bootstrap over mean paired daily active returns, in basis points per day. Wilcoxon tests are paired over daily returns and are reported as diagnostics rather than definitive multiple-comparison-adjusted claims.

Comparison	Mean bp/day	95% CI	Δ Sharpe	Win %	Wilcoxon p
SFP 4-factor vs sentiment-only	0.352	[-1.070, 1.885]	+0.017	27.4	0.556
SCW vs equal-weight SFP	0.182	[-1.224, 1.740]	+0.000	24.8	0.232
Semantic tilt vs price-only forecaster	0.234	[-1.198, 1.497]	+0.021	34.8	0.568
Multi-seed DP-PPO mean vs EW buy-and-hold	-1.392	[-4.119, 1.606]	-0.082	49.7	0.631

511 E.5 SFP Sub-period Performance

512 Table 12 breaks down SFP performance by calendar year and regime window, mirroring Table 7 for
513 DP-PPO. SFP outperforms buy-and-hold in 3 of 5 calendar years (2020: +11pp; 2021: +4pp; 2023:
514 +26pp) and underperforms in 2019 (-8pp) and 2022 (-2pp). The largest single-year outperformance
515 (+26pp in 2023) coincides with the AI-investment-cycle rally that disproportionately benefited
516 NVDA, GOOGL, and AVGO—the high-coverage NASDAQ names that dominate the SFP basket (see
517 SRF/SCW coincidence discussion in Section 5.2). **Caveat:** it is not possible to rule out that SFP’s
518 2023 excess return is a disguised factor exposure to the AI-driven index rally rather than a semantic
519 signal effect; a sector-neutral SFP design would be needed to isolate these contributions, which we
520 leave as future work. Unlike DP-PPO—which outperforms only in the 2020–21 recovery—SFP’s
521 positive excess return is concentrated in the 2020–21 recovery (+39pp) and 2023, both periods of
522 high momentum in the NASDAQ-100 names that dominate the SFP selection basket.

Table 11: SFP transaction-cost sensitivity (2019–2023). Each row re-evaluates the daily top-10 four-factor SFP at the stated per-trade cost. B&H (equal-weight price-average index, same computation as Table 2, 243.6% CR) is cost-free throughout. **SFP is highly turnover-sensitive:** it outperforms B&H only at $\leq 0.1\%$ per trade and underperforms at $\geq 0.2\%$, collapsing at $0.5\%+$. The paper’s main reported result uses 0.1% ; realistic broker costs (0.2 – 0.5%) reverse the advantage. Mirrors Table 8 (DP-PPO tx-cost sensitivity) for the direct portfolio.

Cost (%)	SFP CR (%)	SFP Sharpe	SFP MDD (%)	B&H CR (%)	B&H Sharpe
0.05	363.3	1.151	-35.6	243.6	1.032
0.10	307.2	1.067	-35.6	243.6	1.032
0.20	214.5	0.899	-35.6	243.6	1.032
0.50	44.7	0.395	-59.3	243.6	1.032
1.00	-60.5	-0.436	-86.1	243.6	1.032
2.00	-97.1	-2.002	-98.9	243.6	1.032

Table 12: SFP sub-period performance (2019–2023). Mirrors Table 7 (DP-PPO sub-periods). Excess CR = SFP minus equal-weight B&H (same computation as Table 2). SFP outperforms in 3 of 5 calendar years (2020, 2021, 2023). The single largest annual outperformance (+26pp) occurs in 2023, coinciding with the AI investment cycle rally that disproportionately benefited the high-coverage NASDAQ names (NVDA, GOOGL, AVGO) that dominate the SFP basket; it is not possible to rule out that SFP’s 2023 excess is a disguised factor exposure rather than a semantic signal effect. SFP underperforms in 2019 and 2022 (rate-hike, drawdown year). Cost = 0.1% per trade.

Period	SFP CR (%)	SFP Sharpe	B&H CR (%)	Excess CR (pp)
2019	35.6	1.460	43.3	-7.7
2020	61.9	1.331	51.0	+11.0
2021	46.4	1.827	42.3	+4.1
2022	-30.8	-0.821	-29.0	-1.8
2023	81.5	2.857	55.4	+26.1
2019–20 pre/bear	20.4	0.595	33.8	-13.5
2020–21 recovery	179.0	2.334	139.8	+39.2
2022–23 rate hike	23.3	0.499	9.9	+13.4

E.6 Coverage-Stratified SFP Analysis

Table 3 (presented in Section 5.2) addresses the coverage confound directly. We split the 30-ticker universe into terciles by non-neutral coverage fraction and evaluate SFP within each tercile, comparing against an equal-weight B&H sleeve of the *same tickers*. **Key result: SFP does not outperform equal-weight B&H within any individual coverage tercile.** The full-portfolio outperformance (307.2% vs. 243.6%) reflects a cross-universe *composition* effect—SFP’s top-10 rule preferentially selects mid-coverage names (INTU, KLAC, LRCX, TSLA, ADBE) that happened to deliver strong returns, not names on which the semantic signal is demonstrably more predictive. This result substantially qualifies the main SFP finding; full discussion and the table appear in Section 5.2.

F LLM Signal Validation

Table 13: Descriptive statistics of LLM-generated signal scores (39,995 articles).

	Sentiment	Risk	Confidence	Vol. Forecast
Mean	3.35	2.47	3.51	2.74
Std	1.01	1.02	0.77	0.82
Median	3.00	2.00	3.00	3.00

G Ethics, Deployment, and Reproducibility

This work is a retrospective research benchmark, not a trading recommendation or deployable advisory system. Automated trading systems can amplify losses, liquidity stress, and feedback loops if deployed without governance, risk limits, and human oversight. Our evaluation omits important operational frictions—slippage, partial fills, exchange outages, short-sale constraints, tax effects, and capital limits—so reported results should not be interpreted as expected live performance. The LLM component introduces additional reproducibility concerns: hosted model behaviour can drift over time; prompts may elicit different outputs across providers or versions; and financial news coverage is uneven across firms. Scored article-level signals should be cached and released with prompt templates, model identifiers, timestamps, and dataset hashes. The artifact package provides scripts, processed CSV panels, checkpoints, and dry-run commands for retrained ablations, but full reproducibility still depends on access to the same market data snapshot and LLM-scored signal cache.

LLM Semantic Signal Validation

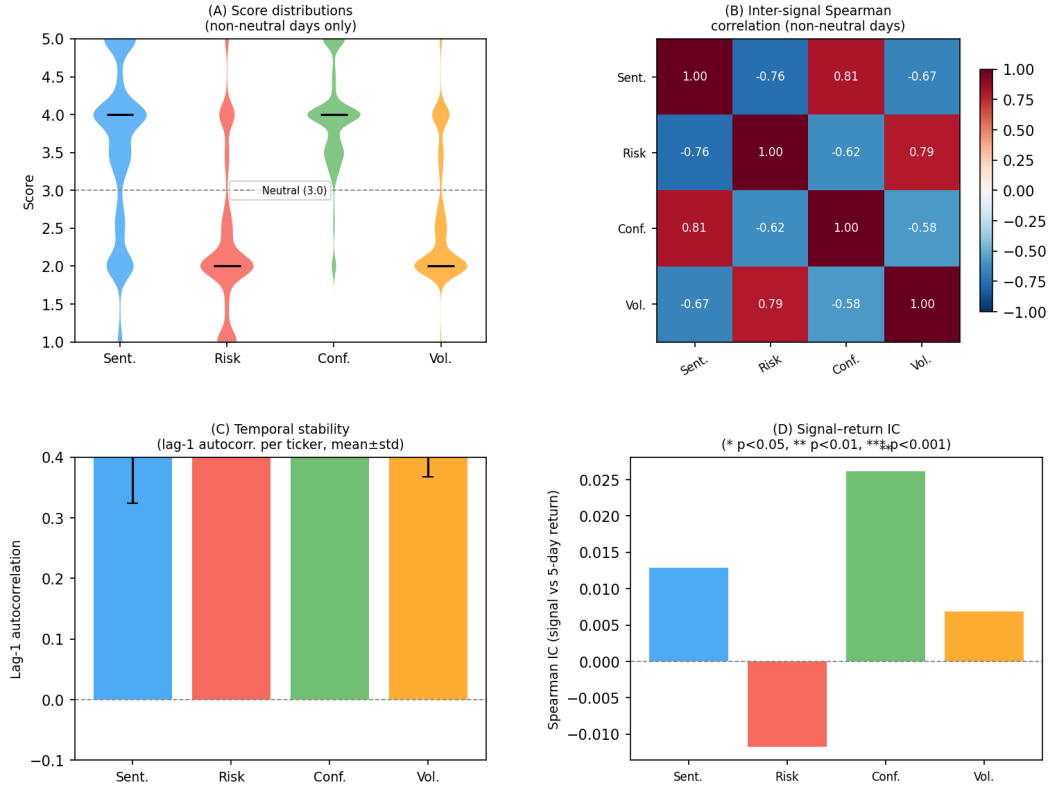


Figure 4: LLM semantic signal validation on the full 2013–2023 panel ($N=83,040$ stock-days, 18,578 non-neutral). (A) Score distributions on non-neutral days show structured directional skew. (B) Inter-signal Spearman correlations replicate expected semantic structure. (C) Lag-1 temporal autocorrelation per ticker confirms scores are not white noise. (D) Signal–return IC: confidence shows a significant positive IC ($p=0.004$).

Table 14: Ticker-level LLM signal coverage on the 2019–2023 trade panel. Coverage is the percentage of stock-days whose score differs from the neutral imputation value 3.0; rows show the five most-covered and five least-covered tickers.

Ticker	Stock-days	Any (%)	Sent.	Risk	Conf.	Vol.
GOOGL	1258	26.9	22.1	23.4	19.0	22.4
AVGO	1258	26.6	19.8	24.4	16.6	23.1
NVDA	1258	26.5	22.1	23.4	19.0	20.5
MU	1258	24.6	21.2	22.8	18.0	21.4
ASML	1258	23.4	17.9	19.5	13.3	16.1
INTC	1258	0.5	0.5	0.5	0.5	0.5
META	1258	0.0	0.0	0.0	0.0	0.0
MSFT	1258	0.0	0.0	0.0	0.0	0.0
ISRG	1258	0.0	0.0	0.0	0.0	0.0
COST	1258	0.0	0.0	0.0	0.0	0.0

Table 15: Signal validity proxy: pooled Spearman information coefficients between LLM signal coordinates and 5-day forward returns / absolute returns on the 2019–2023 stock-date panel. This is not a human annotation audit, but it checks whether scores align with realized market quantities rather than pure noise.

Signal	Return IC	p	Return IC	p
sentiment	0.0044	0.39	-0.0047	0.36
risk	-0.0152	0.0032	0.0122	0.018
confidence	0.0217	2.5e-05	0.0065	0.21
volatility_forecast	-0.0045	0.38	0.0175	0.00068

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: Abstract and Section 1 state SSAI contributions and diagnostic RL framing; Section 6 ties claims to reported experiments.

Guidelines:

- The answer NA means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A No or NA answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Section 6 (Limitations and future work) and ethics Appendix G discuss coverage, masking caveats, seed power, and deployment gaps.

Guidelines:

- The answer NA means that the paper has no limitation while the answer No means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.

- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [NA]

Justification: No formal theorems or lemmas; Definition 1 (Appendix A) specifies an interface formally but without asymptotic theory.

Guidelines:

- The answer NA means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Appendix B (data pipeline, hyperparameters, prompts), anonymized supplementary scripts, and checkpoint conventions document how to replicate evaluations.

Guidelines:

- The answer NA means that the paper does not include experiments.
- If the paper includes experiments, a No answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.

- (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
- (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
- (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Supplementary/code artifact includes README, requirements.txt, evaluation harnesses, and processed CSV panels; LLM re-scoring may require API access.

Guidelines:

- The answer NA means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://nips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so “No” is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://nips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental Setting/Details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: Train/test windows (2013–2018 / 2019–2023), portfolio rules, PPO/SAC settings, and signal construction appear in Section 3 and Appendix B.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment Statistical Significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Multi-seed means $\pm$ std, bootstrap Sharpe intervals, Wilcoxon diagnostics, and paired block-bootstrap tables (e.g., Table 10); variability sources stated in text.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The authors should answer "Yes" if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g. negative error rates).
- If error bars are reported in tables or plots, The authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments Compute Resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix B (Compute) reports hardware, LLM batch runtime, and per-run RL training time.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code Of Ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: Authors reviewed the NeurIPS Code of Ethics; retrospective benchmark without human subjects.

Guidelines:

- The answer NA means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer No, they should explain the special circumstances that require a deviation from the Code of Ethics.

- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader Impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Appendix G discusses misuse risks of automated trading and omitted frictions.

Guidelines:

- The answer NA means that there is no societal impact of the work performed.
- If the authors answer NA or No, they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [NA]

Justification: We do not release a standalone high-risk foundation model; LLM access is via third-party API and prompts/checkpoints are documentation-focused.

Guidelines:

- The answer NA means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: FinRL fork, FNSPID/Yahoo-derived panels, and libraries are cited; HF dataset identifier given (license per dataset card).

Guidelines:

- The answer NA means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New Assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [\[Yes\]](#)

Justification: Anonymized supplementary zip documents scripts, checkpoints, and signal CSV conventions alongside the paper.

Guidelines:

- The answer NA means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and Research with Human Subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [\[NA\]](#)

Justification: No crowdsourcing or human-participant experiments.

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects

847 Question: Does the paper describe potential risks incurred by study participants, whether
848 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
849 approvals (or an equivalent approval/review based on the requirements of your country or
850 institution) were obtained?

851 Answer: [NA]

852 Justification: No human subjects research.

853 Guidelines:

- 854 • The answer NA means that the paper does not involve crowdsourcing nor research with
855 human subjects.
- 856 • Depending on the country in which research is conducted, IRB approval (or equivalent)
857 may be required for any human subjects research. If you obtained IRB approval, you
858 should clearly state this in the paper.
- 859 • We recognize that the procedures for this may vary significantly between institutions
860 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
861 guidelines for their institution.
- 862 • For initial submissions, do not include any information that would break anonymity (if
863 applicable), such as the institution conducting the review.

864 16. Declaration of LLM usage

865 Question: Does the paper describe the usage of LLMs if it is an important, original, or
866 non-standard component of the core methods in this research?

867 Answer: [Yes]

868 Justification: Section 3 and Appendix B specify fixed-prompt multi-axis LLM scoring via a
869 hosted API as core state construction.

870 Guidelines:

- 871 • The answer NA means that the core method development in this research does not
872 involve LLMs as any important, original, or non-standard components.
- 873 • Please refer to the NeurIPS Main Track Handbook for what should or should not be
874 described.